

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      /* CMP-135 SDTM DM Creation
71      Domain: DM*/;
72
73      options validvarname=upcase nofmterr dlcreatedir;
74
75      libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
NOTE: Libref RAW was successfully assigned as follows:
Engine:          V9
Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1
76      libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";
NOTE: Libref SDTM was successfully assigned as follows:
Engine:          V9
Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm
77
78      libname valxpt xport
79      "/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/dm.xpt";
NOTE: Libref VALXPT was successfully assigned as follows:
Engine:          XPORT
Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/dm.xpt
80
81
82
83      %macro make_date(invar=, outdt=);
84      &outdt = .;
85
86      if vtype(&invar) = "N" then do;
87      if not missing(&invar) then do;
88      if abs(&invar) > 50000 then &outdt = datepart(&invar);
89      else &outdt = &invar;
90      end;
91      end;
92      else do;
93      _dtm = input(strip(&invar), anydtdtm.);
94      if not missing(_dtm) then &outdt = datepart(_dtm);
95
96      if missing(&outdt) then
97      &outdt = input(strip(&invar), anydtdte.);
98      end;
99      %mend;
100
101      %macro make_dtm(datevar=, timevar=, outdt=, outdtm=);
102      %make_date(invar=&datevar, outdt=&outdt);
103
104      _tm = .;
105
106      if vtype(&timevar) = "N" then _tm = &timevar;
107      else _tm = input(strip(&timevar), time.);
108
109      if not missing(&outdt) and not missing(_tm) then
110      &outdtm = dhms(&outdt, hour(_tm), minute(_tm), second(_tm));
111      else if not missing(&outdt) then
112      &outdtm = dhms(&outdt, 0, 0, 0);
113      %mend;
114
115      data ex_prep;
116      set raw.ex;
117
118      %make_dtm(datevar=exstdtn, timevar=exsttm, outdt=exdt, outdtm=exdtm);
119
120      format exdt date9. exdtm datetime20.;
121      run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).
118:1 118:1

NOTE: Character values have been converted to numeric values at the places given by: (Line):(Column).
118:2

NOTE: There were 1904 observations read from the data set RAW.EX.

NOTE: The data set WORK.EX_PREP has 1904 observations and 36 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	2691.65k
OS Memory	25768.00k
Timestamp	06/29/2026 07:23:12 PM
Step Count	436 Switch Count 2
Page Faults	0
Page Reclaims	316
Page Swaps	0

```

Voluntary Context Switches      17
Involuntary Context Switches    0
Block Input Operations           0
Block Output Operations          2056

```

```

122
123      proc sql;
124          create table ex_dates as
125          select
126              subject,
127              min(exdt)  as first_dose_dt  format=date9.,
128              max(exdt)  as last_dose_dt   format=date9.,
129              min(exdtm) as first_dose_dtm format=datetime20.,
130              max(exdtm) as last_dose_dtm  format=datetime20.
131          from ex_prep
132          where not missing(exdt)
133          group by subject;

```

NOTE: Table WORK.EX_DATES created, with 24 rows and 5 columns.

```

134      quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              6675.12k
      OS Memory           30632.00k
      Timestamp           06/29/2026 07:23:13 PM
      Step Count          437  Switch Count  5
      Page Faults         0
      Page Reclaims       345
      Page Swaps          0
      Voluntary Context Switches 17
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 272

```

```

135
136      data ds_prep;
137          set raw.ds;
138
139          %make_date(invar=dsstdat, outdt=dsdt);
140
141          format dsdt date9.;
142      run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

139:1 139:1

NOTE: There were 24 observations read from the data set RAW.DS.

NOTE: The data set WORK.DS_PREP has 24 observations and 25 variables.

```

NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.01 seconds
      memory              1130.03k
      OS Memory           24996.00k
      Timestamp           06/29/2026 07:23:13 PM
      Step Count          438  Switch Count  2
      Page Faults         0
      Page Reclaims       83
      Page Swaps          0
      Voluntary Context Switches 18
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264

```

```

143
144      proc sql;
145          create table ds_dates as
146          select
147              subject,
148              max(dsdt) as eos_dt format=date9.
149          from ds_prep
150          where not missing(dsdt)
151          group by subject;
NOTE: Table WORK.DS_DATES created, with 24 rows and 2 columns.

```

```

152      quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds

```

```

system cpu time    0.00 seconds
memory            5867.28k
OS Memory         30120.00k
Timestamp         06/29/2026 07:23:13 PM
Step Count        439   Switch Count   2
Page Faults       0
Page Reclaims     150
Page Swaps        0
Voluntary Context Switches 11
Involuntary Context Switches 1
Block Input Operations 0
Block Output Operations 272

```

```

153
154
155
156     data enr_dates;
157         set raw.enr;
158
159         %make_date(invar=cnstdtn, outdt=consent_dt);
160
161         format consent_dt date9.;
162
163         keep subject consent_dt enrgrp;
164     run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

159:1 159:1

NOTE: There were 24 observations read from the data set RAW.ENR.

NOTE: The data set WORK.ENR_DATES has 24 observations and 3 variables.

NOTE: DATA statement used (Total process time):

```

real time        0.00 seconds
user cpu time    0.00 seconds
system cpu time  0.00 seconds
memory          1106.84k
OS Memory       24996.00k
Timestamp       06/29/2026 07:23:13 PM
Step Count      440   Switch Count   2
Page Faults     0
Page Reclaims   87
Page Swaps      0
Voluntary Context Switches 22
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 264

```

```

165
166
167     data inv_prep;
168         set raw.inv;
169
170         length siteid $4 invnam $13;
171
172         siteid = strip(siteid);
173         invnam = catx(" ", invfname, invlname);
174
175         keep siteid invid invnam country;
176     run;

```

WARNING: In a call to the CATX function, the buffer allocated for the result was not long enough to contain the concatenation of all the arguments. The correct result would contain 14 characters, but the actual result might either be truncated to 13 character(s) or be completely blank, depending on the calling environment. The following note indicates the left-most argument that caused truncation.

NOTE: Argument 3 to function CATX(' ','Gerard '[12 of 30 characters shown],'Vockley '[12 of 30 characters shown]) at line 173 column 14 is invalid.

PROJECT=CMP135 INVLNAME=Vockley INVFNNAME=Gerard INVID=0133 SITEID=0125 COUNTRY=USA INVNAM= _ERROR_=1 _N_=5

NOTE: There were 12 observations read from the data set RAW.INV.

NOTE: The data set WORK.INV_PREP has 12 observations and 4 variables.

NOTE: DATA statement used (Total process time):

```

real time        0.00 seconds
user cpu time    0.00 seconds
system cpu time  0.00 seconds
memory          1056.75k
OS Memory       24996.00k
Timestamp       06/29/2026 07:23:13 PM
Step Count      441   Switch Count   2
Page Faults     0
Page Reclaims   88
Page Swaps      0
Voluntary Context Switches 17
Involuntary Context Switches 0

```

```

Block Input Operations      0
Block Output Operations     264

```

```

177
178
179      data dov_prep;
180          set raw.dov;
181
182          %make_date(invar=visdtn, outdt=dmdt);
183
184          folder_u = upcase(strip(foldername));
185          inst_u   = upcase(strip(instancename));
186
187          format dmdt date9.;
188      run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

182:1 182:1

NOTE: There were 644 observations read from the data set RAW.DOV.

NOTE: The data set WORK.DOV_PREP has 644 observations and 26 variables.

NOTE: DATA statement used (Total process time):

```

real time      0.00 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         1459.93k
OS Memory      25256.00k
Timestamp      06/29/2026 07:23:13 PM
Step Count     442  Switch Count  2
Page Faults    0
Page Reclaims  108
Page Swaps     0
Voluntary Context Switches  18
Involuntary Context Switches 0
Block Input Operations      0
Block Output Operations     776

```

```

189
190      proc sql;
191          create table dov_screen as
192          select
193              subject,
194              min(dmdt) as dmdt format=date9.
195          from dov_prep
196          where index(folder_u, "SCREEN") > 0
197              or index(inst_u, "SCREEN") > 0
198          group by subject;

```

NOTE: Table WORK.DOV_SCREEN created, with 24 rows and 2 columns.

```

199      quit;

```

NOTE: PROCEDURE SQL used (Total process time):

```

real time      0.00 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         6150.18k
OS Memory      30248.00k
Timestamp      06/29/2026 07:23:13 PM
Step Count     443  Switch Count  2
Page Faults    0
Page Reclaims  182
Page Swaps     0
Voluntary Context Switches  11
Involuntary Context Switches 1
Block Input Operations      0
Block Output Operations     272

```

```

200
201
202      proc sql;
203          create table dm_merge as
204          select
205              a.*,
206              b.first_dose_dt,
207              b.last_dose_dt,
208              b.first_dose_dtm,
209              b.last_dose_dtm,
210              c.eos_dt,
211              d.consent_dt,
212              d.enrgrp,
213              f.invid,
214              f.invnam,

```

```

215         f.country,
216         g.dmdt
217     from raw.dm as a
218     left join ex_dates   as b on a.subject = b.subject
219     left join ds_dates   as c on a.subject = c.subject
220     left join enr_dates  as d on a.subject = d.subject
221     left join inv_prep   as f on scan(a.subject, 1, "-") = f.siteid
222     left join dov_screen as g on a.subject = g.subject;
NOTE: Table WORK.DM_MERGE created, with 24 rows and 40 columns.

```

```

223     quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      user cpu time       0.00 seconds
      system cpu time     0.01 seconds
      memory              59020.75k
      OS Memory           82384.00k
      Timestamp           06/29/2026 07:23:13 PM
      Step Count          444   Switch Count   2
      Page Faults         0
      Page Reclaims       1151
      Page Swaps           0
      Voluntary Context Switches  46
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 288

```

```

224
225
226     data sdtm.dm;
227         retain
228             STUDYID DOMAIN SUBJID RFSTDTC RFENDTC RFXSTDTC RFXENDTC
229             RFICDTC RFPENDTC DTHDTC DTHFL SITEID INVID INVNAM BRTHDTC
230             AGE AGEU SEX RACE ETHNIC ARM ACTARM COUNTRY DMDTC DMDY
231             USUBJID ACTARMCD ARMCD;
232
233         length
234             STUDYID   $6
235             DOMAIN    $2
236             SUBJID    $9
237             RFSTDTC   $10
238             RFENDTC   $10
239             RFXSTDTC  $16
240             RFXENDTC  $16
241             RFICDTC   $10
242             RFPENDTC  $10
243             DTHDTC    $1
244             DTHFL     $1
245             SITEID    $4
246             INVID     $4
247             INVNAM    $13
248             BRTHDTC   $10
249             AGEU      $5
250             SEX        $1
251             RACE       $5
252             ETHNIC     $22
253             ARM        $7
254             ACTARM     $7
255             COUNTRY    $3
256             DMDTC     $10
257             USUBJID   $40
258             ACTARMCD  $8
259             ARMCD     $8;
260
261     set dm_merge;
262
263     label
264         STUDYID = "Study Identifier"
265         DOMAIN  = "Domain Abbreviation"
266         SUBJID  = "Subject Identifier for the Study"
267         RFSTDTC = "Subject Reference Start Date/Time"
268         RFENDTC = "Subject Reference End Date/Time"
269         RFXSTDTC = "Date/Time of First Study Treatment"
270         RFXENDTC = "Date/Time of Last Study Treatment"
271         RFICDTC = "Date/Time of Informed Consent"
272         RFPENDTC = "Date/Time of End of Participation"
273         DTHDTC  = "Date/Time of Death"
274         DTHFL   = "Subject Death Flag"
275         SITEID  = "Study Site Identifier"
276         INVID   = "Investigator Identifier"
277         INVNAM  = "Investigator Name"
278         BRTHDTC = "Date/Time of Birth"

```

```

279         AGE      = "Age"
280         AGEU      = "Age Units"
281         SEX       = "Sex"
282         RACE      = "Race"
283         ETHNIC    = "Ethnicity"
284         ARMCD     = "Planned Arm Code"
285         ARM       = "Description of Planned Arm"
286         ACTARMCD  = "Actual Arm Code"
287         ACTARM    = "Description of Actual Arm"
288         COUNTRY   = "Country"
289         DMDTC     = "Date/Time of Collection"
290         DMDY      = "Study Day of Collection";
291 /*         USUBJID = "Unique Subject Identifier"; */
292
293         STUDYID = "CMP135";
294         DOMAIN  = "DM";
295
296         SUBJID  = strip(subject);
297         USUBJID = catx("-", STUDYID, SUBJID);
298
299         RFSTDTC = put(first_dose_dt, yymmdd10.);
300         RFENDTC = put(eos_dt, yymmdd10.);
301         RFPENDTC = RFENDTC;
302
303         if not missing(first_dose_dtm) then
304             RFXSTDTC = cats(
305                 put(datepart(first_dose_dtm), yymmdd10.),
306                 "T",
307                 put(timepart(first_dose_dtm), tod5.)
308             );
309
310         if not missing(last_dose_dtm) then do;
311             if timepart(last_dose_dtm) = 0 then
312                 RFXENDTC = put(datepart(last_dose_dtm), yymmdd10.);
313             else
314                 RFXENDTC = cats(
315                     put(datepart(last_dose_dtm), yymmdd10.),
316                     "T",
317                     put(timepart(last_dose_dtm), tod5.)
318                 );
319         end;
320
321         RFICDTC = put(consent_dt, yymmdd10.);
322
323         DTHDTC = "";
324         DTHFL  = "";
325
326         SITEID = scan(strip(subject), 1, "-");
327         INVID  = strip(invid);
328         INVNAM = strip(invnam);
329
330         %make_date(invar=brthdtn, outdt=brthdt);
331
332         BRTHDTC = put(brthdt, yymmdd10.);
333
334         if not missing(brthdt) and not missing(consent_dt) then
335             AGE = int(yrdif(brthdt, consent_dt, "ACTUAL"));
336         else AGE = agecalc;
337
338         AGEU = "YEARS";
339
340         SEX   = strip(sex_cod);
341         RACE  = upcase(strip(race));
342         ETHNIC = upcase(strip(ethnic));
343
344         ARMCD   = "CMP135_5";
345         ARM     = "Group 1";
346         ACTARMCD = "CMP135_5";
347         ACTARM  = "Group 1";
348
349         COUNTRY = strip(country);
350
351         DMDTC = put(dmdt, yymmdd10.);
352
353         if not missing(dmdt) and not missing(first_dose_dt) then do;
354             if dmdt >= first_dose_dt then DMDY = dmdt - first_dose_dt + 1;
355             else DMDY = dmdt - first_dose_dt;
356         end;
357
358         keep
359             STUDYID DOMAIN SUBJID RFSTDTC RFENDTC RFXSTDTC RFXENDTC
360             RFICDTC RFPENDTC DTHDTC DTHFL SITEID INVID INVNAM BRTHDTC
361             AGE AGEU SEX RACE ETHNIC ARM ACTARM COUNTRY DMDTC DMDY

```

```

362          USUBJID ACTARMCD ARMCD;
363      run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

330:1 330:1

NOTE: Variable AGECALC is uninitialized.

WARNING: Multiple lengths were specified for the variable SEX by input data set(s). This can cause truncation of data.

WARNING: Multiple lengths were specified for the variable RACE by input data set(s). This can cause truncation of data.

WARNING: Multiple lengths were specified for the variable INVID by input data set(s). This can cause truncation of data.

NOTE: There were 24 observations read from the data set WORK.DM_MERGE.

NOTE: The data set SDTM.DM has 24 observations and 28 variables.

NOTE: DATA statement used (Total process time):

real time	0.02 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	1323.75k
OS Memory	27048.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	445 Switch Count 1
Page Faults	0
Page Reclaims	119
Page Swaps	0
Voluntary Context Switches	34
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	280

```

364
365
366      proc sort data=sdm.dm out=dm_created;
367          by STUDYID USUBJID;
368      run;

```

NOTE: There were 24 observations read from the data set SDTM.DM.

NOTE: The data set WORK.DM_CREATED has 24 observations and 28 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1200.56k
OS Memory	27308.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	446 Switch Count 2
Page Faults	0
Page Reclaims	145
Page Swaps	0
Voluntary Context Switches	23
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

```

369
370      proc sort data=valxpt.dm out=dm_validate;
371          by STUDYID USUBJID;
372      run;

```

NOTE: There were 24 observations read from the data set VALXPT.DM.

NOTE: The data set WORK.DM_VALIDATE has 24 observations and 28 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.01 seconds
memory	18187.42k
OS Memory	43204.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	447 Switch Count 2
Page Faults	0
Page Reclaims	360
Page Swaps	0
Voluntary Context Switches	28
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

373
374
375      ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DM_PROC_COMPARE.rtf";
NOTE: Writing RTF Body file: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/DM_PROC_COMPARE.rtf
376
377      title "PROC COMPARE: SDTM DM Created vs Validation DM";

```

```

378
379     proc compare base=dm_validate compare=dm_created listall criterion=0.000001;
380         id STUDYID USUBJID;
381     run;

```

NOTE: There were 24 observations read from the data set WORK.DM_VALIDATE.

NOTE: There were 24 observations read from the data set WORK.DM_CREATED.

NOTE: PROCEDURE COMPARE used (Total process time):

real time	0.01 seconds
user cpu time	0.02 seconds
system cpu time	0.00 seconds
memory	2239.15k
OS Memory	27564.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	448 Switch Count 0
Page Faults	0
Page Reclaims	317
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	8

```

382
383     ods rtf close;
384
385     title "Created DM";
386     proc print data=dm_created;
387 run;

```

NOTE: There were 24 observations read from the data set WORK.DM_CREATED.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.07 seconds
user cpu time	0.08 seconds
system cpu time	0.01 seconds
memory	1035.93k
OS Memory	26532.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	449 Switch Count 0
Page Faults	0
Page Reclaims	65
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	48

```

388
389     title "Validation DM";
390     proc print data=dm_validate(obs=3);
391 run;

```

NOTE: There were 3 observations read from the data set WORK.DM_VALIDATE.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.01 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	871.00k
OS Memory	26532.00k
Timestamp	06/29/2026 07:23:13 PM
Step Count	450 Switch Count 0
Page Faults	0
Page Reclaims	62
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	16

```

392
393     title;
394
395     OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
405

```